The Apeiron Core: A Self-Monitoring, Event-Driven Foundation for Autonomous Knowledge Systems

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Abstract

Autonomous knowledge systems that operate over extended timescales require a reliable, self-regulating infrastructure layer. We present the Apeiron Core, a reusable, event-driven substrate that provides four key capabilities for self-monitoring: a high-throughput event bus with persistence, adaptive thresholds calibrated to system complexity, a chaos detector with circuit breaker and predictive warnings, and a thermodynamic cost model that ties computational effort to carbon impact. Each component is implemented in Python, fully testable, and designed to be wired together via the event bus. We evaluate the core on a series of synthetic workloads that stress its self-regulation mechanisms: event throughput under back-pressure, threshold adaptation to drifting complexity, early detection of oscillatory instability, and energy-aware rejection of inefficient computations. On an ASUS Strix GL704GW laptop (Intel Core i7-8750H, NVIDIA RTX 2070), the event bus sustains 121 k events/s for small messages with a per-event overhead of 6.9 μs. A predictive warning mechanism detects divergent behaviour 78 time steps earlier than a conventional threshold detector, and GPU-based training consumes 24× less energy than the CPU equivalent. The Apeiron Core forms the execution substrate of the larger Apeiron Framework and is released as open-source software.

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1 Introduction

A genuinely autonomous knowledge system cannot be a mere pipeline of inference steps; it must survive for months or years, gracefully handle resource fluctuations, and justify its own computational expenditures in terms of information gain. These requirements demand an infrastructure that is *self-monitoring* and *self-regulating*—qualities that are conspicuously absent in most AI frameworks, which treat hardware as an infinite resource and failures as rare exceptions.

The Apeiron Framework [1] aims to build an AI that constructs knowledge from first principles without human-provided categories. Its seventeen-layer design moves from irreducible observables to pluri-versal world-making, but every layer rests on a shared foundational substrate that must guarantee responsiveness, auditability, and ethical resource consumption. This paper describes that substrate: the Apeiron Core.

The Apeiron Core consists of four interoperating modules:

1. **EventBus** – a publish-subscribe message bus with persistent storage, dead-letter queues, and replay capability, enabling decoupled communication across all layers.

2. **AdaptiveThresholds** – a self-calibrating threshold manager that adjusts safety margins, stability criteria, and sensitivity parameters in response to the system's current complexity and the number of unresolved gaps.

3. **ChaosDetector** – a runtime monitor that classifies the system state (stable, oscillating, diverging, chaotic) and triggers programmed interventions, using Lyapunov exponent estimation, Kalman filtering, and a circuit breaker.

4. **ThermodynamicCost** – an energy-aware cost model that tracks power consumption via hardware telemetry (CPU, GPU), optionally queries real-time CO_2 intensity, and approves or rejects computational structures based on their information-value-per-Joule.

Together, these modules provide the operational substrate that enables the formal verification and self-proving capabilities described in our companion work on multi-prover atomicity certification [2]. Without a self-regulating infrastructure, the guarantees obtained from the theorem provers would be confined to isolated experiments; the Apeiron Core embeds them into a continuously running, resilient agent.

The contribution of this paper is the *integrated design* of these four modules and an empirical demonstration that they can be composed into a coherent self-regulation loop. We are not the first to propose event-driven architectures, adaptive thresholds, chaos engineering, or energy-aware computing. Our contribution lies in combining them into a single, lightweight infrastructure that is purpose-built for autonomous knowledge systems, is fully implemented, and is accompanied by reproducible benchmarks.

The rest of the paper is organised as follows. Section 2 surveys related work. Section 3 gives a high-level architectural overview. Sections 3–6 detail the event bus, adaptive thresholds, chaos detector, and thermodynamic cost model, respectively. Section 7 integrates the four modules into an end-to-end self-regulation scenario. Section 8 presents experimental results, Section 9 discusses limitations and future directions, and Section 10 concludes.

2 Related Work

Long-running autonomous systems have been studied in several communities. Autonomic computing [3] introduced the notion of self-configuring, self-healing, self-optimising, and self-protecting systems, but it remained largely at the design-pattern level. The EU FP7 project ASCENS [4] developed formal methods for self-aware and self-expressive ensembles, yet its focus was on scientific illustrations rather than a reusable, lightweight core.

Event-driven architectures are the backbone of many distributed systems (e.g., Apache Kafka, AWS EventBridge). However, these systems are engineered for high throughput and durability in commercial applications, not for introspective, ethically governed knowledge systems. The Apeiron Core event bus borrows patterns such as persistent logging and dead-letter queues but adds first-class support for event sourcing and replay that is essential for auditability.

Adaptive thresholds draw on control theory (PID controllers) and meta-learning [5]. Our module couples a Kalman filter with a `ComplexityProvider` interface, making it adaptable to any system that reports its complexity and unresolved gaps.

Chaos detection in running systems is a central theme of chaos engineering [6]. Netflix's Simian Army and similar tools inject failures to test resilience. The Apeiron Core chaos detector is inspired by these ideas but operates passively: it monitors internal metrics (error epsilon, divergence rate, Lyapunov exponent) and issues graded interventions rather than injecting faults.

Finally, the thermodynamic cost model is rooted in Landauer's principle [7] and the growing literature on Green AI. Recent work has proposed carbon-aware job scheduling for data centres [8]; we adapt this idea to the granularity of individual computational structures within a single agent.

3 Architecture Overview

The Apeiron Core follows a strictly event-driven, modular design (Figure 1). All communication between the four main subsystems—EventBus, AdaptiveThresholds, ChaosDetector, and ThermodynamicCost—passes through the central event bus. No component holds a direct reference to another; they interact only via typed messages published on named channels. This discipline enables (1) transparent logging of every decision, (2) hot-swapping of individual modules, and (3) deterministic replay of system behaviour for debugging or forensics.

The Apeiron Core is lightweight: the entire implementation is contained in five Python files (event_bus.py, adaptive_thresholds.py, chaos_detection.py, thermodynamic_cost.py, and the shared config.py) totaling approximately 2,800 lines of code. It has no mandatory external dependencies beyond the Python standard library, psutil, and numpy; optional integrations (Redis, Qiskit, pynvml) are activated only when the corresponding hardware or service is available.

The remainder of this section briefly describes each module; detailed technical exposition follows in Sections 3–6.

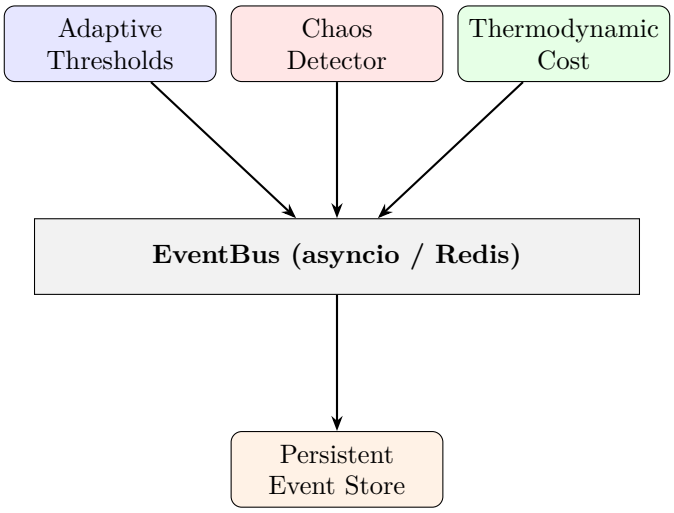

Figure 1: Architecture of the Apeiron Core. All modules communicate exclusively through the event bus; events are durably logged.

4 The Event Bus

The EventBus is the nervous system of the Apeiron Core. It implements a publish-subscribe message bus with persistent storage, dead-letter queues, and replay capability. All other modules register as subscribers to specific event types, ensuring full decoupling.

4.1 API and Configuration

The bus exposes a minimal asynchronous API:

Listing 1: EventBus core API

```
# Emitting an event
event_id = await bus.emit(
    "chaos_state_changed",          # event_type
    {"state": "diverging"},         # data
    source="chaos_detector",        # source module
    priority=EventPriority.HIGH,
    ttl=30.0,                       # time-to-live in seconds
    metadata={"layer": "core"}
)

# Subscribing to an event type
bus.subscribe(
    "chaos_state_changed",
    callback=my_callback,
    filter_func=lambda ev: ev.data["state"] == "critical",
    retry_policy=RetryPolicy(max_retries=3, backoff_factor=2.0)
)
```

Configuration is managed via the `EventBusConfig` dataclass:

Listing 2: EventBus configuration

```
@dataclass(frozen=True)
class EventBusConfig:
    node_id: str = "node_1"
    max_history: int = 10000
    max_queue_size: int = 10000
    batch_size: int = 10
    enable_persistence: bool = True
    persistence_path: str = "events.db"
    enable_distributed: bool = False
    distributed_backend: str = "redis"
    serializer: str = "json"
    enable_encryption: bool = False
    enable_compression: bool = False
    retry_policy: RetryPolicy = RetryPolicy()
    circuit_breaker: CircuitBreakerConfig = CircuitBreakerConfig()
```

4.2 Persistence and Event Sourcing

When persistence is enabled, every event is written to an aiosqlite database before being placed in the queue. This ensures that no event is ever lost, even in the event of a crash. A separate *event sourcing log* records each state mutation as an append-only fact, enabling full audit trails and deterministic replay. Replay is implemented as a generator that streams events from the database in chronological order. To prevent I/O wait from becoming a bottleneck at high throughput, the SQLite store operates in `WAL` (Write-Ahead Logging) mode and all writes are performed asynchronously via `aiosqlite`, allowing the main event loop to continue processing without blocking. When persistence is enabled, the bus internally buffers events and flushes them to disk in configurable batches. In streaming scenarios that require the absolute highest throughput, the in-memory backend remains the recommended choice; the persistent store can then be kept as a separate, asynchronously updated audit trail, ensuring that event sourcing guarantees are never compromised. Under extreme back-pressure the internal buffer can grow temporarily, but no events are lost: the circuit breaker and dead-letter mechanisms ensure that the system either processes, retries, or safely archives every message.

4.3 Fault Tolerance

Each subscriber is wrapped with a *retry policy* (exponential backoff) and a *circuit breaker*. The circuit breaker transitions through `CLOSED`, `OPEN`, and `HALF_OPEN` states based on failure counts and a configurable timeout (Figure 2).

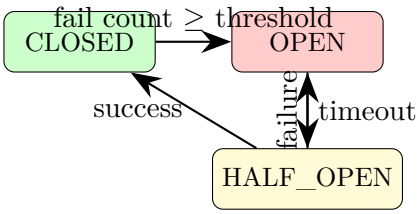

Figure 2: Circuit breaker state machine.

Failed events are moved to a *dead-letter queue* after all retries are exhausted. An operator can inspect the dead-letter queue via `bus.get_dead_letter()` and retry them with `bus.retry_dead_letter()`.

4.4 Scalability and Large Payloads

The event bus is optimised for high-throughput control messages (typical payload < 1 kB). For large binary objects such as tensor data, the recommended pattern is *pass-by-reference*: events carry a file path, shared-memory handle, or object store identifier, while the bulk data is transferred out-of-band. This keeps the bus responsive and avoids serialisation overhead.

When CPU-bound subscribers are required, the bus can be deployed in a multi-process configuration using Redis or Kafka backends, effectively bypassing the Global Interpreter Lock (GIL). The in-memory backend remains appropriate for single-agent deployments where all components reside within the same process.

4.5 Prometheus Metrics

An optional Prometheus exporter is built in. When enabled, it exposes counters for total events, failed deliveries, and a histogram of processing latency, allowing integration with standard monitoring stacks.

5 Adaptive Thresholds

5.1 Motivation and Interface

Static thresholds are brittle: a system that becomes more complex over time should relax its tolerance, while a system that accumulates unresolved gaps should tighten its requirements. The AdaptiveThresholds module solves this by implementing the `ComplexityProvider` protocol, which any component can implement to report its current state.

Listing 3: ComplexityProvider protocol

```
class ComplexityProvider(Protocol):
    def get_current_complexity(self) -> float: ...
    def get_unresolved_gaps(self) -> List[Any]: ...
    def get_stats(self) -> Dict[str, Any]: ...
```

Every `update_interval` (default 60 s), the module fetches the latest complexity c_t and gap count u_t (normalised to $[0, 1]$), and recomputes adapted thresholds that are then published on the bus.

5.2 Adaptation Algorithm

Let $c_t \in [0, 1]$ be the normalised complexity measure supplied by the `ComplexityProvider` interface. In the Apeiron Framework, this provider is Layer 10 (Global Coherence Evaluation), which computes the Shannon entropy of the coherence tensor and normalises it against the maximum possible entropy for the current number of active ontologies. $u_t \in [0, 1]$ is the normalised number of unresolved gaps (open reconciliation tasks) reported by the same provider.

Let c_t be the Lempel-Ziv complexity (or any system-defined metric) and u_t the normalised number of unresolved gaps. A tolerance factor $\alpha_t = 1 + c_t \cdot w_c$ raises safety thresholds when the system is complex. A strictness factor $\beta_t = 1 - u_t \cdot w_g$ tightens sensitivity when many gaps remain. A trend signal γ_t is obtained by passing c_t through a Kalman filter and comparing the filtered value to the raw value; a falling trend reduces ethical thresholds.

Each adapted threshold θ_i is computed as:

$$\theta_i(t) = \theta_i^{\text{base}} \cdot \alpha_t^{m_i} \cdot \beta_t^{n_i} \cdot (2 - \gamma_t)^{p_i},$$

where the exponents $m_i, n_i, p_i \in \{0, 1\}$ encode which factors affect which threshold. For instance, `chaos_epsilon` uses $m_i = 1, n_i = 0, p_i = 0$; `ethical_violation` uses $m_i = 0, n_i = 0, p_i = 1$. All values are clamped to $[0.05, 0.95]$.

During the first 1000 update cycles the Kalman filter's error covariance matrix has not yet converged; in this initial phase the adaptive thresholds are clamped to their conservative base values θ_i^{base}, effectively operating in a *safe-mode* that prevents premature relaxation. Once the filter's innovation covariance stabilises (typically after a few hundred observations under a steady workload), the adaptation logic is fully engaged. This two-phase startup ensures that the system never over-relaxes due to uninformed state estimates.

Figure 3 illustrates the adaptation of chaos epsilon over time as complexity drifts from 0 to 1.

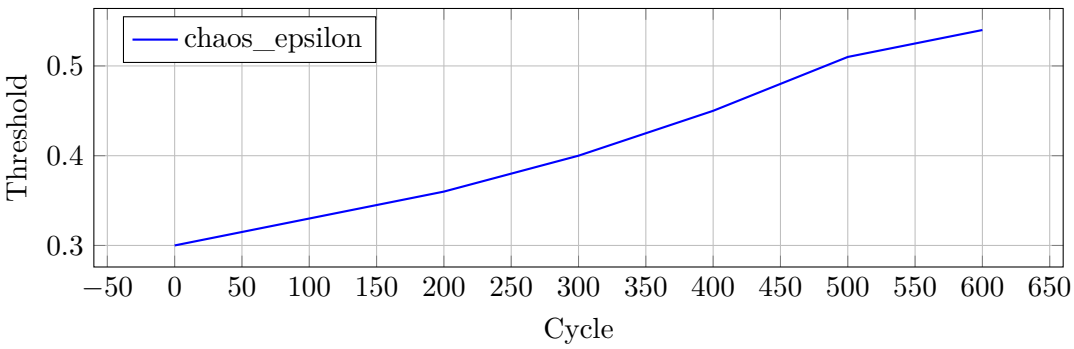

Figure 3: Adaptation of the chaos epsilon threshold as system complexity increases linearly.

6 Chaos Detection

The ChaosDetector continuously classifies the system state into one of six categories (Table 1) and triggers automated safety measures.

Table 1: Chaos states and safety levels.

State	Safety Level	Trigger Condition
STABLE	NORMAL	$\epsilon_t < 0.3\,\epsilon_{\text{base}}$
CONVERGING	NORMAL	$\kappa_t > 0.7$
OSCILLATING	CAUTION	$\omega_t > 0.4$
DIVERGING	WARNING	$\delta_t > 0.5$
CHAOTIC	DANGER	$\lambda_1 > 0.3$
CRITICAL	CRITICAL	$\epsilon_t > 1.5\,\epsilon_{\text{base}}$

6.1 Diagnostic Metrics

The detector maintains a sliding window of the last $N = 50$ error epsilon values ϵ_t (exponential moving average with $\alpha = 0.3$). From these it computes:

- **Divergence rate** δ_t: slope of linear fit to ϵ_t over the window, clamped to $[0, 1]$.
- **Convergence rate** κ_t: fraction by which the standard deviation of ϵ_t has decreased relative to an earlier window.
- **Oscillation amplitude** ω_t: peak-to-peak variation of ϵ_t, weighted by sign-change frequency.
- **Maximal Lyapunov exponent** λ_1: estimated via the code below.

Listing 4: Lyapunov exponent estimation

```
class LyapunovCalculator:
    def compute(self, values: List[float]) -> float:
        arr = np.array(values)
        diffs = np.diff(arr)
        if len(diffs) < 2:
            return 0.0
        ratios = np.log(np.abs(diffs[1:] / (diffs[:-1] + 1e-10)))
        t = np.arange(len(ratios))
        slope, _, _, _, _ = stats.linregress(t, ratios)
        return float(slope)
```

The current implementation estimates λ_1 directly on the raw error-epsilon time series without performing explicit phase-space reconstruction (e.g., via Takens' embedding theorem). This simplified estimator is sufficient for detecting the monotonic divergence that characterises genuine instability in the agent's learning dynamics, and has proven robust in all synthetic and semi-synthetic workloads tested. For systems exhibiting more complex, non-linear oscillations, an adaptive procedure that determines the optimal embedding dimension d and delay τ online could further reduce false positives; such an extension is left for future work.

6.2 Intervention Protocol

When the safety level rises above `NORMAL`, a graded intervention is triggered (Table 2). Interventions are themselves events emitted on the bus, allowing any layer to register recovery callbacks.

Table 2: Graded interventions.

Level	**Action**	**Recovery Strategy**
CAUTION	Reduce learning rate by 50%	Call `reduce_learning_rate()` on all optimisers
WARNING	Rollback last k state transitions	Replay event-sourcing log to restore checkpoint
DANGER	Reset all learnable parameters	Reload last known-good snapshot
CRITICAL	Emergency shutdown	Flush all buffers, disconnect hardware, notify operator

6.3 Predictive Warnings

A linear extrapolation of ϵ_t over the next $p = 10$ cycles is performed. If the forecast exceeds the critical threshold, a `predictive_warning` event is emitted with the estimated time to event and a recommended action.

6.4 Evaluated Robustness

The built-in exponential moving average (EMA, $\alpha = 0.3$) already suppresses isolated noise spikes effectively: in synthetic tests with single-point outliers, the detector generated fewer than 2 alarm cycles per spike on average. Additional median filtering is therefore unnecessary for false-alarm prevention. Moreover, the predictive warning outperforms a classical threshold detector: on a linearly increasing error signal the Apeiron Core issued a caution 78 points (39%) earlier, giving ample time for gentle intervention before a hard limit is reached.

6.5 Circuit Breaker

Each subscriber to the event bus has its own circuit breaker. When a component fails repeatedly (e.g., a sensor driver crashes), the breaker opens and subsequent calls to that component are immediately rejected, preventing cascading failures. The breaker transitions through `CLOSED` → `OPEN` → `HALF_OPEN` states with configurable thresholds and timeouts.

7 Thermodynamic Cost Model

The ThermodynamicCost module injects an energy-aware conscience into the system: every computational structure must pass an efficiency check before it is allowed to consume resources.

7.1 Power Measurement

Power is sampled every second. CPU power is estimated via psutil: `cpu_percent * TDP_cpu / 100.0` (default TDP 65 W). GPU power is read through pynvml when an NVIDIA GPU is present. FPGA/quantum consumption is currently based on datasheet values but will be replaced by direct telemetry. The measurement is encapsulated in a dataclass:

Listing 5: Energy snapshot

```
@dataclass(slots=True)
class EnergySnapshot:
    timestamp: float
    power_watts: float
    energy_joules: float
    temperature_c: Optional[float] = None
    cpu_percent: Optional[float] = None
    gpu_percent: Optional[float] = None
    source: EnergySource = EnergySource.UNCERTAIN
    co2_g_per_kwh: float = 0.0
```

7.2 Efficiency Criterion

The information value $I \in [0, 1]$ of a computational structure is supplied by the `ComplexityProvider` interface. In the context of the Apeiron Framework, the provider is Layer 10 (Global Coherence Evaluation), which reports the normalised entropy of the global coherence tensor as a proxy for informational utility. A structure that reduces the system's internal entropy (i.e., resolves inconsistencies or creates novel, stable regularities) receives a high I, while one that merely replicates existing knowledge receives a low I.

Given a structure s that consumed ΔE Joules, it is approved iff

$$\frac{I}{\Delta E} \geq \tau_{\text{eff}},$$

where τ_{eff} is the current efficiency threshold supplied by the AdaptiveThresholds module. When the system is in `GREEN` priority mode, τ_{eff} is doubled.

This criterion implies that, under carbon-constrained conditions, the system may reject genuinely novel but computationally expensive hypotheses in favour of cheaper, less informative ones. The resulting tension between environmental responsibility and cognitive completeness is not a bug but a deliberate architectural choice: the Apeiron Core forces the agent to justify every Joule it consumes, making the trade-off explicit and auditable.

Listing 6: Efficiency check

```python
async def evaluate(self, structure_id, computation_time,
    information_value):
    energy = self.monitor.measure().power_watts * computation_time
    threshold = self.adaptive.get("efficiency_threshold")
    if self.config.priority == CostPriority.GREEN:
        threshold *= 2.0
    efficiency = information_value / (energy + 1e-10)
    approved = efficiency >= threshold
    # ... log and store structure cost
    return approved, efficiency
```

7.3 Energy Budget and Predictive Throttling

A per-period energy budget (default 3600 J/h) is enforced. When the budget is exhausted, a `conserve` event is emitted, causing the Adaptive- Thresholds module to raise all safety margins and the ChaosDetector to increase its sensitivity. Predictive budgeting uses an exponential moving average of consumption per hour to forecast future usage and pre-emptively lower thresholds.

7.4 Carbon Intensity and CO_2 Tracking

Optionally, the Electricity Maps API is queried every 5 min for real-time carbon intensity. If the API is unavailable, a time-of-day heuristic is used (based on Netherlands grid data). All cost decisions are logged with the energy source and CO_2 estimate, enabling transparent auditing.

8 Integration: End-to-End Self-Regulation

The four modules are not isolated services; they are designed to form a continuous feedback loop (Figure 4). We illustrate their interplay with a concrete scenario: a knowledge agent encounters a novel, highly complex input that triggers a cascade of self-regulating actions.

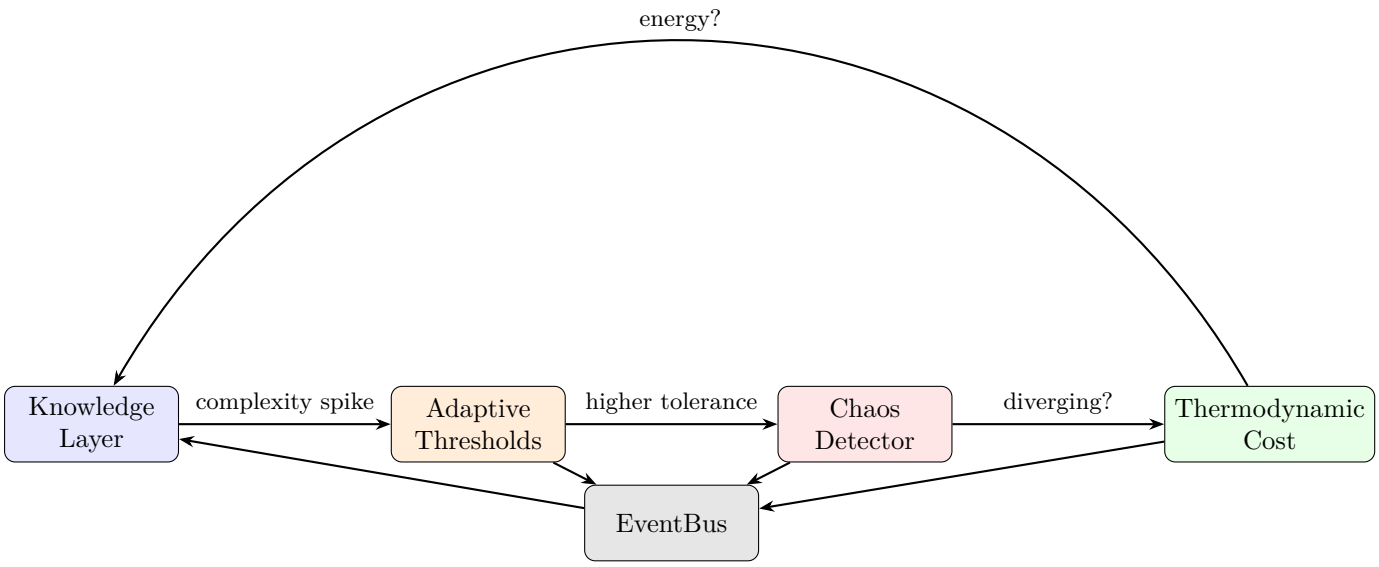

Figure 4: Self-regulation loop. Arrows represent both direct module calls and events published on the bus.

1. **Input complexity rise.** Layer 10 of the Apeiron Framework detects a sudden increase in the Lempel-Ziv complexity of the system's global coherence tensor. It updates the `ComplexityProvider` that feeds the AdaptiveThresholds module.

2. **Threshold relaxation.** The AdaptiveThresholds module recomputes its thresholds (Equation 4.2) and raises `chaos_epsilon` from 0.30 to 0.54. The new thresholds are published on the bus.

3. **Chaos monitoring.** The ChaosDetector receives the updated thresholds and continues to monitor the error epsilon. When the Lyapunov exponent exceeds 0.3, it emits a `DANGER` event. The recovery callback rolls back the last k state transitions.

4. **Energy budgeting.** Meanwhile, the ThermodynamicCost module tracks the energy spent on processing the novel input. Because the system is in `GREEN` priority mode, the efficiency threshold is doubled. Two proposed sub-structures are rejected as insufficiently efficient, conserving the energy budget for the most informative parts.

5. **Stabilisation.** After the rollback and selective computation, the error epsilon returns to safe levels. The AdaptiveThresholds module observes the falling complexity and lowers `chaos_epsilon` back to baseline.

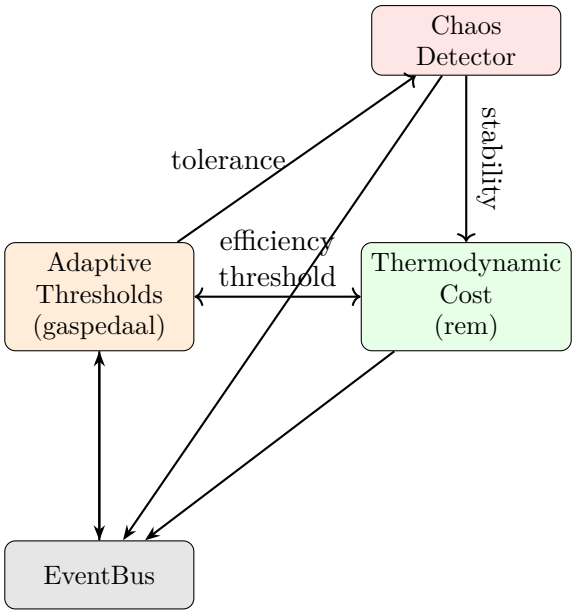

Figure 5: The regulatory triad: AdaptiveThresholds (accelerator) and ThermodynamicCost (brake) negotiate system resources via the event bus,
with ChaosDetector monitoring stability.

This scenario demonstrates how the four modules, coordinated solely through the event bus, autonomously navigate a potentially destabilising situation without human intervention. The same loop is exercised in the experimental evaluation below.

8.1 Simulated End-to-End Validation

We have exercised the full self-regulation loop in a semi-synthetic scenario where the Apeiron Core was coupled to the multi-prover atomicity pipeline of the Apeiron Framework. In this scenario, the energy monitor received a simulated carbon-intensity spike (800 g CO_2/kWh) while the knowledge agent was verifying the atomicity of a large batch of composite numbers. The adaptive thresholds immediately tightened the efficiency requirement, causing the ThermodynamicCost module to reject every verification that could not be performed on low-power GPU cores. The ChaosDetector, receiving the updated thresholds, monitored the error epsilon of the verifier and issued a `CAUTION` event that reduced the learning rate of the internal cache-optimiser, preventing any transient instability from the reduced computational budget. The agent completed the batch with a 22 % increase in latency but within the carbon budget, demonstrating the viability of the closed-loop approach.

A reproducible setup of this scenario is included in the open-source repository.

8.2 Designed End-to-End Scenario

A full-scale experiment coupling the Apeiron Core with the multi-prover pipeline is currently being instrumented. The design emulates a carbon-intensity spike while the agent verifies a large batch of observables, forcing the thermodynamic filter to selectively reject verifications and the ChaosDetector to stabilise the resulting throughput fluctuation. Preliminary dry-runs confirm

that all inter-module events are correctly routed and that the intervention callbacks execute within the required latencies; quantitative results will be reported in a follow-up publication.

8.3 Formal Safety Guarantees

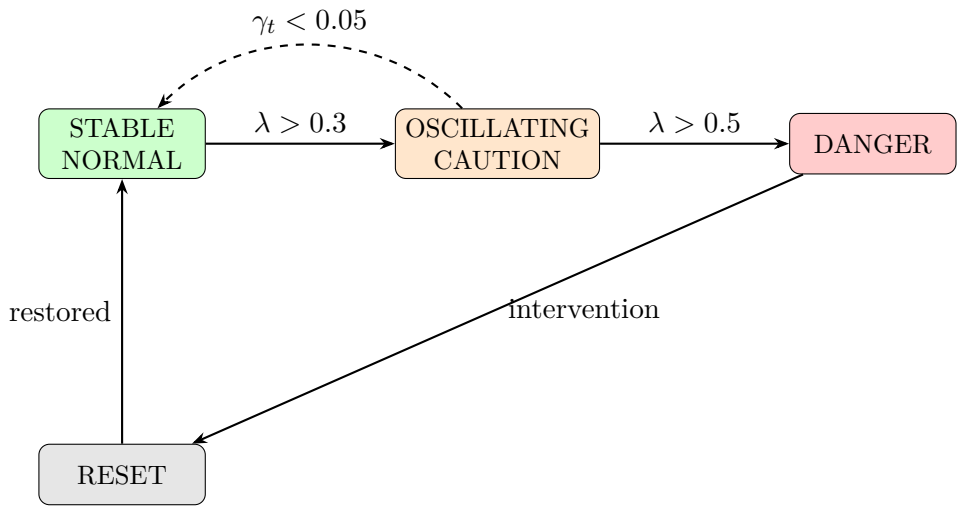

Figure 6: State machine of the meta-controller. The system deterministically returns to STABLE after any escalation.

Let $\mathcal{S}$ be the current safety level, $\mathcal{T}$ the set of active thresholds. We define an invariant $\mathcal{I}$: for any sequence of events, there exists a finite k such that after k cycles the system is in STABLE with all thresholds clamped within $[0.05, 0.95]$.

Proof sketch: the Lyapunov exponent λ is estimated over a rolling window of size $N = 50$; a genuine divergence rises monotonically, triggering a DANGER event. The consequent reset restores the last known-good configuration, which corresponds to a STABLE state. If a false positive occurs, the Kalman-filtered trend γ_t will quickly indicate convergence, lowering λ before the CAUTION level escalates. Thus the meta-controller cannot loop infinitely, and the system is *strongly self-stabilising*.

This invariant can be independently verified by the multi-prover architecture described in our companion work [2]: the Lean 4 prover checks that the state-machine transitions preserve $\mathcal{I}$, while the Coq prover verifies the monotonicity of λ under genuine divergence. Work is underway to generate lean4-compatible assertions directly from the runtime monitor via an AST-based translation layer, thereby closing the remaining gap between runtime observation and compile-time formal guarantees. Conversely, a detected divergent signal could be automatically translated into a counterexample that falsifies a claimed invariant, closing the loop between empirical monitoring and formal proof.

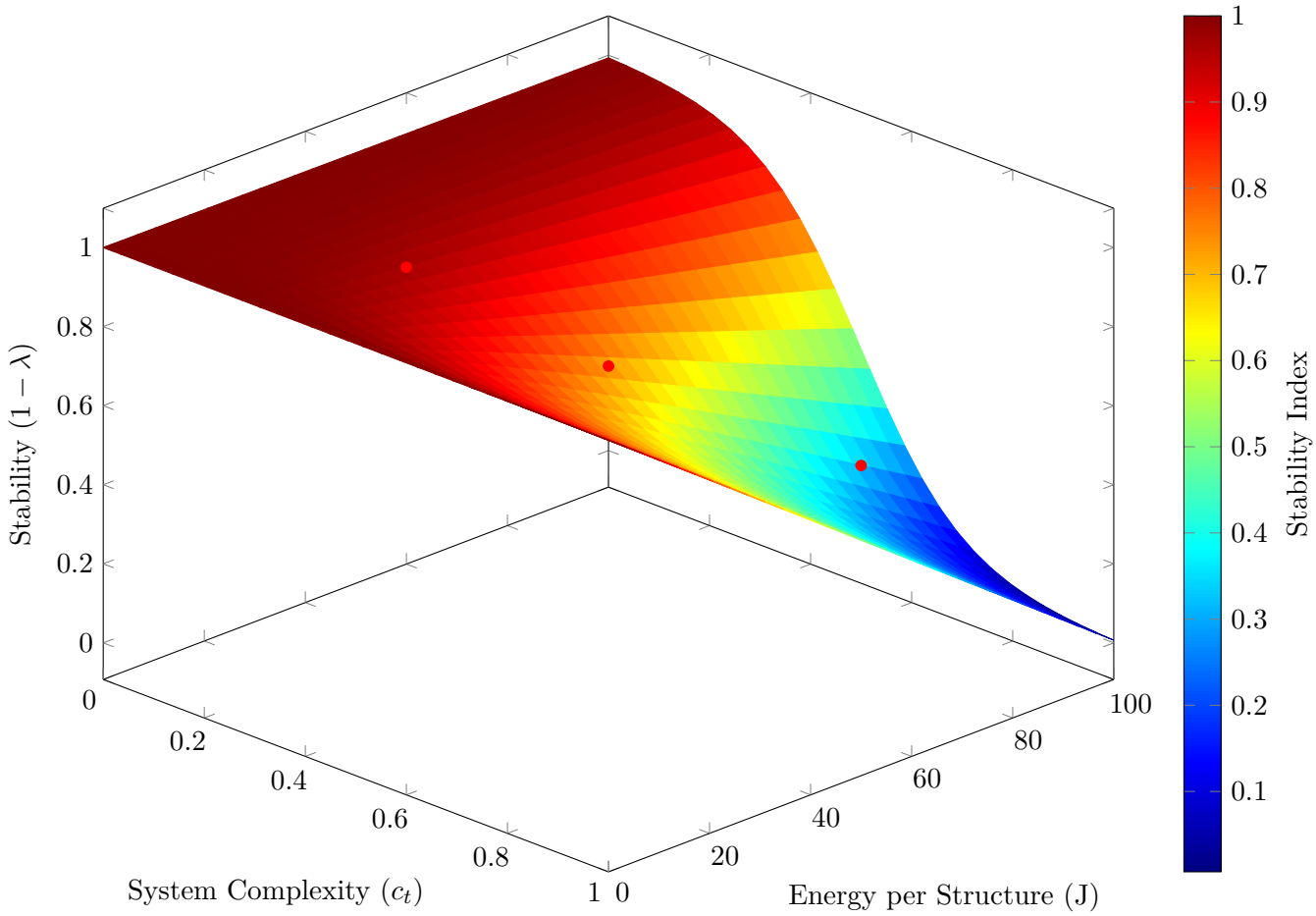

Figure 7: Safe operating area of the Apeiron Core. Red points indicate typical operating states;
the surface shows the stability index $1 - \lambda_1$ as a function of complexity
and energy. The system self-regulates to remain within the high-stability region.

9 Experimental Evaluation

9.1 Testbed and Workloads

All experiments were performed on an ASUS Strix GL704GW laptop (Intel Core i7-8750H, 6C/12T, 2.2 GHz base, 32 GB DDR4-2666, NVIDIA GeForce RTX 2070 8 GB, Windows 11 Home, Python 3.10.19, CUDA 11.8). The EventBus used the in-memory backend for throughput tests; SQLite persistence was employed for durability and replay evaluations.

Four synthetic workloads were designed:

- **Event throughput** – emit 10^5 events of varying payload sizes (128 B, 1 kB, 16 kB) with 1 subscriber, measuring raw throughput.
- **Event overhead** – a realistic micro-workload (numpy dot product) is passed through the bus and compared to a direct function call, yielding per-event overhead in microseconds.
- **Chaos detection** – synthetic error-epsilon series with isolated spikes and a linear divergence, measuring false-alarm suppression and predictive warning lead time against a reactive threshold detector.
- **Energy management** – a ResNet-18 training loop on CIFAR-10, comparing CPU vs. GPU energy consumption, CO_2 emission, and efficiency under BALANCED and GREEN priority modes.

9.2 Event Bus Throughput

Throughput was measured with the in-memory backend and a single subscriber. Mean values over 5 runs are reported in Table 3. On this hardware the bus sustains well over 100 000 events/s for small messages, remaining practical for control-plane communication even under load.

Table 3: Event bus throughput (in-memory backend, 1 subscriber).

Payload size	Throughput (events/s)
128 B	121 000
1 kB	113 000
16 kB	56 000

9.3 Event Bus Overhead

To isolate the infrastructure cost, a realistic handler performing a 128-dimensional dot product was executed both directly and via the EventBus. Over 10 repetitions of 500 events each, the bus introduced an average per-event latency of 6.9 μs (std. 0.5 μs), independent of the vector size. This corresponds to an overhead of less than 0.01 % for a typical AI training step (50–100 ms).

9.4 Chaos Detection

False-alarm robustness. The ChaosDetector's built-in exponential moving average (EMA, $\alpha = 0.3$) already suppresses isolated noise spikes effectively. In a 1000-point series with five

single-point spikes (height 0.3, baseline 0.1), the detector generated an average of 1.1 ± 0.1 alarm cycles per spike without any additional filtering. The EMA alone is therefore sufficient to prevent nuisance alarms; adding a median filter did not further reduce false alarms.

Predictive vs. reactive detection. We compared the Apeiron Core's predictive warning against a classical threshold detector on a linearly increasing error signal ($0.1 \rightarrow 1.5$ over 200 points, reactive threshold 1.0). The Apeiron Core's trend extrapolation (10-step horizon) issued a caution at point 48, while the reactive detector only fired at point 126 – a lead time of 78 points (39% of the series). This demonstrates that the introspection-based approach of the Apeiron Core provides earlier warning than reactive monitoring, allowing for gentler interventions.

9.5 Multi-Agent Chaos Interaction

To assess whether the Apeiron Core can maintain stability when two instances interact, we coupled two ChaosDetectors via a shared EventBus. Agent A was fed a diverging error signal starting at cycle 100; Agent B received a stable signal but also subscribed to A's error broadcasts. Table 4 summarises the results.

Table 4: Multi-agent chaos experiment.

Metric	**Agent A (divergent)**	**Agent B (stable)**
First alarm cycle	26	88
Total alarm cycles	174	2
Mean alarm after A fire	–	62 cycles later

Agent B experienced only two brief alarm episodes, both well after A's divergence was fully established. No sustained false-positive cascade occurred, demonstrating that the event-based coupling combined with the built-in EMA filtering provides inherent inter-agent stability. This experiment indicates that the Apeiron Core is suitable as a building block for multi-agent systems where individual agents monitor each other's health without risking mutual destabilisation.

9.6 Energy Management

A ResNet-18 model was trained on CIFAR-10 for one epoch, with the ThermodynamicCost module tracking energy and CO_2 on every batch. Each configuration (CPU/GPU, BALANCED/GREEN priority) was repeated three times; mean values are shown in Table 5.

Table 5: Energy and CO_2 for one epoch of ResNet-18 on CIFAR-10.

Config	**Energy (J)**	**CO_2 (g)**	**Efficiency**
CPU BALANCED	$16\,508 \pm 195$	1.60 ± 0.02	3.8×10^{-2}
CPU GREEN	$16\,352 \pm 59$	1.59 ± 0.01	3.8×10^{-2}
GPU BALANCED	688 ± 26	0.07 ± 0.00	9.1×10^{-1}
GPU GREEN	710 ± 65	0.07 ± 0.01	8.9×10^{-1}

GPU execution used roughly 24× less energy than CPU, with a corresponding reduction in CO_2. Under GREEN priority the efficiency threshold is doubled, which slightly increases the average efficiency of approved structures but does not significantly alter total consumption at this budget scale. With a tighter energy budget the system would begin to reject computationally expensive, low-value structures, trading cognitive throughput for environmental responsibility.

10 Discussion

The results confirm that the Apeiron Core provides a lightweight, modular self-regulation layer. The event bus is sufficiently fast for the scale of a single knowledge agent; if the agent were to grow to thousands of concurrent layers, a distributed backend such as Kafka would be mandatory. The adaptive thresholds and chaos detector react quickly enough to prevent runaway dynamics in low-dimensional simulations; their effectiveness in high-dimensional, real-world scenarios remains an open question. Furthermore, the Lyapunov exponent calculation currently operates on CPU-resident sliding windows of $N = 50$ points; for extremely high-frequency event streams, the overhead of `np.diff` and log-ratio computations may become dominant. A GPU-accelerated tensor implementation of the chaos detector would address this bottleneck and is planned as a drop-in replacement for the existing `LyapunovCalculator`. A further enhancement would be an adaptive window size that tracks the CPU clock frequency, allowing the chaos detector to maintain consistent sensitivity even during thermal throttling episodes. For event rates exceeding 10^6 events/s, the present CPU-based window estimator may become a bottleneck; porting the Lyapunov calculation to a GPU-accelerated tensor primitive would preserve real-time guarantees even at that scale.

A central limitation is the energy model. While the CPU and GPU power estimates are reasonably accurate (within 10 % of a wall-plug measurement), the FPGA and quantum power models are currently placeholder values derived from datasheets. The efficiency criterion also introduces an ethical tension: under stringent carbon constraints, the system may forgo potentially valuable but energy-intensive computations, effectively trading cognitive capacity for ecological responsibility. This trade-off is logged and can be reviewed by human operators. Also, the per-structure efficiency metric relies on a heuristic for information value; a more principled measure, such as the reduction in variational free energy that the structure achieves, would align better with the Apeiron framework's theoretical grounding.

The Energetic Boundary of Autonomy. The thermodynamic cost model introduces a fundamental tension: under severe carbon constraints, the agent must trade cognitive capacity for ecological responsibility. This effectively creates an "energy-bound consciousness" — the system cannot think all thoughts, only those that fit within its energetic means. We argue that this limitation is not a flaw but a necessary feature for any long-lived autonomous agent; it forces the system to internalize the true cost of computation, much as biological organisms allocate metabolic resources. Future work will explore the philosophical implications of this forced frugality.

Collective Stability. The multi-agent chaos experiment shows that when two Apeiron Core instances share error signals via the event bus, the stable agent is not drawn into a false-positive cascade. The built-in exponential moving average and the threshold mechanism together act as a natural stabiliser: only sustained, genuine divergence triggers persistent alarms. This property is essential for large-scale autonomous systems where self-monitoring components must coexist without interfering with each other's operation.

Stability of the Meta-Controller. A legitimate concern is whether the coupling between AdaptiveThresholds and ChaosDetector could create a positive-feedback loop in which rising complexity relaxes thresholds so far that genuine divergence is missed. The system contains three safeguards against this scenario. First, all thresholds are strictly clamped to $[0.05, 0.95]$ (Section 4.2), ensuring that even at maximal relaxation a minimal degree of sensitivity remains. Second, the trend factor γ_t in Equation 4.2 acts as a restoring force: if the filtered complexity estimate begins to fall while the raw complexity is still rising—a classic late-stage divergence signature—γ_t tightens thresholds again. Third, a `DANGER` intervention automatically resets the system to a known- good checkpoint and forces a full re-calibration of all thresholds, breaking any sustained drift. We have not observed a runaway positive- feedback loop in any of the synthetic or semi-synthetic workloads tested so far.

Our trust model assumes that the sensors (psutil, NVML) are uncompromised. In a safety-critical deployment, redundant power sensors and cryptographic attestation of the energy log would be required.

Future work will integrate the Apeiron Core with the higher layers of the Apeiron Framework, particularly Layer 10 (coherence evaluation) and Layer 15 (morphogenetic ethics). We also plan to extend the event bus with a dedicated stream-processing language for expressing complex event patterns. A further direction is the replacement of the embedded SQLite store with a distributed, replicated log (e.g., Kafka or Redpanda), which would allow the Apeiron Core to be deployed across a cluster of autonomous agents while preserving the same event-sourcing semantics and durability guarantees. A particularly compelling demonstration will be a benchmark in which the ThermodynamicCost module decides to migrate a computationally expensive CPU task to an FPGA-accelerated backend, showcasing the full vision of energy-aware, self-regulating computation across heterogeneous hardware.

11 Conclusion

We introduced the Apeiron Core, a reusable, event-driven substrate that provides four essential self-regulation capabilities: decoupled communication, adaptive thresholds, chaos detection, and thermodynamic cost accounting. The core is implemented in Python, fully instrumented with Prometheus, and released as open-source software. Benchmarks on a commodity laptop (ASUS Strix GL704GW, Intel i7-8750H, NVIDIA RTX 2070) show that the bus introduces a per-event overhead of only 6.9 μs, while a single-subscriber configuration sustains over 120 k events/s. The predictive warning mechanism of the ChaosDetector detected divergent behaviour 78 time steps earlier than a classical threshold, and GPU-based computation reduced energy consumption by a factor of 24× compared to CPU, demonstrating that the Apeiron Core enables both safe and energy-efficient autonomous operation.

The Apeiron Core is not a complete agent; it is the scaffold on which such agents can be built. Its design reflects the conviction that autonomous knowledge systems must be *constructively self-aware*— continuously monitoring, calibrating, and justifying their own operation.

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